

Bioremediation of Wastewater – Effect of Algae in Bioremediation of Nitrate and Phosphate Content in Wastewater

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ABSTRACT: The progressive adoption of an urban, industrial lifestyle has deteriorated the quality of freshwater reservoirs and caused a global challenge for satisfying the demand and maintaining water quality. Industrial, agricultural and domestic processes contaminate water with chemical and biological pollutants that cannot be released into the environment until treated. Traditional wastewater treatment plants are effective in remediating suspended solids and many harmful elements, but they are not always effective in nitrate and phosphate removal causing them to be discharged into the environment. This study aims to determine the effectiveness of algae to bioremediate nitrate and phosphate content from wastewater. Four algae species were tested with wastewater in a bioreactor setup for ten days. The nitrate and phosphate contents were measured every day using a HACH-DR-890 colorimeter to see how effectively the specimen remediated the nitrate and phosphate contents. The analysis shows all four species were able to effectively remediate 93-96% of nitrate and 73-86% of phosphate. *Chlamydomonas reinhardtii* was the most effective in nitrate remediation with a 95.66% remediation and *Scenedesmus quadricauda* was the least effective with 93.97% remediation. For phosphates, *S. quadricauda* (85.22%) was most the effective with 85.22% remediation and *A. platensis* was the least effective with a 73.36% remediation. This study concludes that algae bioremediation is viable in treating nitrates and phosphates in wastewater in a natural, sustainable way compared to conventional treatment processes.

KEYWORDS: Environment pollution; Water treatment; Wastewater; Bioremediation; Bioreactor; Algae; Nitrate; Phosphate.

■ INTRODUCTION

Maintaining the quality of water is a global challenge. The growing population along with the progressive adoption of an urban, industrial lifestyle has deteriorated the quality of freshwater reservoirs around the world. Remediation of wastewater generated from domestic sewage, industrial, and agricultural discharges has become a large concern for both developed and developing countries. Wastewater contains physical, chemical, and biological pollutants including harmful substances that cannot be released into the environment until the wastewater is treated¹ per the EPA guidelines.

Traditional wastewater treatment plants are effective in the remediation of suspended solids but toxic heavy metals, like arsenic and mercury, and nitrate and phosphate content removal processes are not always very effective and sustainable, causing these toxic elements to be discharged into ground water. Bioremediation is an effective and eco-friendly method in removing those elements. It uses naturally occurring microorganisms to break down hazardous substances into less toxic or nontoxic substances.² This study's objective is to determine the effectiveness of algae bioremediation on the nitrate and phosphate content in wastewater. Algae absorbs nitrates and phosphates and can clean the water by separating the nutrients as biomass. This algal biomass can also be used later as an energy source and turned into biofuel, which is renewable and has a small carbon footprint.

■ RESULTS AND DISCUSSION

Based on the analysis, it is observed that all four types of algae were able to reduce the nitrate concentration from wastewater. The initial concentration of nitrate in the wastewater sample ranged from 178.25 - 188.50 mg/L, an average of 183.28 mg/L. The four algae types were able to reduce it to the following: *Chlamydomonas reinhardtii* (average: 7.96 mg/L), *Arthrospira platensis* (average: 9.96 mg/L), *Scenedesmus quadricauda* (average: 11.06 mg/L) and *Chlorella vulgaris* (average: 9.20 mg/L) as shown in Table 1.

Table 1. Nitrate concentration over 10 days (Average of 5 trials).

Days	Control Solution	Sp-1 (C. reinhardtii)	Sp-2 (A. platensis)	Sp-3 (S. quadricauda)	Sp-4 (C. vulgaris)
0	183.28	183.28	183.28	183.28	183.28
1	177.79	143.62	153.90	160.96	166.48
2	170.20	101.87	127.38	130.30	141.55
3	164.33	76.84	88.97	103.83	98.41
4	159.48	47.14	62.13	79.77	72.56
5	155.05	26.83	35.45	53.79	36.99
6	152.53	19.32	21.96	34.74	20.91
7	149.61	13.49	16.94	22.77	16.69
8	146.13	11.19	13.27	16.21	12.82
9	142.90	10.03	11.93	13.51	10.49
10	140.05	7.96	9.96	11.06	9.20

Based on this data, the nitrate remediation efficiency of the algae specimens was calculated. The average percentage nitrate

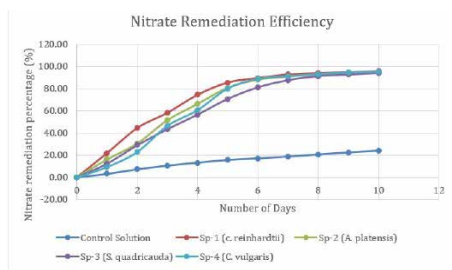


Figure 1. Nitrate remediation efficiency (%) of the algae specimens.

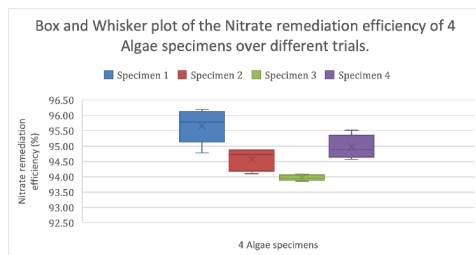


Figure 2. Phosphate remediation efficiency (%) of the algae specimens.

removal achieved by the four specimens were *C. reinhardtii* (95.66%), *A. platensis* (94.57%), *S. quadricauda* (93.97%) and *C. vulgaris* (94.98%) as shown in Fig. 1. Statistical analysis was performed to determine the different remediation efficiencies over the trials. The box and whisker plot shown in Fig. 2 shows the nitrate remediation efficiencies of the four algae species. It can be concluded *C. reinhardtii* is able to achieve the highest nitrate removal with 95.66%. All the specimens were able to remove 93-96% of wastewater nitrates. The EPA's nitrate concentration limit in water is 10 mg/L. All but one algal specimen was able to bring the nitrate concentration to that level over the ten-day study.

Table 2. Phosphate concentration over 10 days (Average of 5 trials)

Days	Control Solution	Sp-1 (<i>C. reinhardtii</i>)	Sp-2 (<i>A. platensis</i>)	Sp-3 (<i>S. quadricauda</i>)	Sp-4 (<i>C. vulgaris</i>)
0	24.77	24.77	24.77	24.77	24.77
1	24.08	20.55	22.32	21.72	20.74
2	23.29	18.44	19.80	19.11	17.31
3	22.39	18.31	16.77	17.28	14.70
4	21.83	12.08	14.71	14.29	12.05
5	21.32	10.55	12.48	11.63	10.73
6	20.23	9.02	10.46	9.86	9.10
7	19.48	7.87	8.46	7.90	7.17
8	18.93	6.86	7.71	6.47	5.94
9	18.23	6.18	7.03	5.21	4.95
10	17.14	5.42	6.60	3.86	4.34

For the phosphate remediation, all four algal species efficiently removed phosphorus from the wastewater. The phosphate concentration remediation by each algae specimen is presented in Table 2. Prior to algae treatment the initial phosphorus content was, on average, 24.77 mg/L. After ten days the phosphorus concentration was reduced to no more than 3.66 mg/L.

Based on this data, the phosphate remediation efficiency by *S. quadricauda* was the greatest during the experimental period. 85.22% of the initial phosphate content was consumed by *S. quadricauda*. It was then followed by *C. vulgaris* (82.46%)

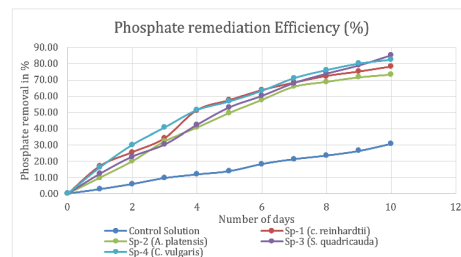


Figure 3. Phosphate remediation efficiency (%) of the algae specimens.

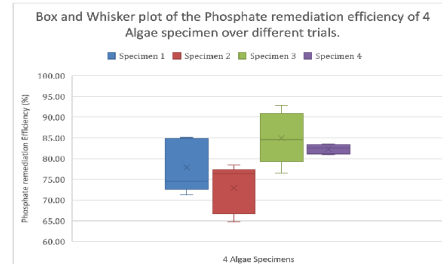


Figure 4. Statistical analysis of Phosphate remediation efficiency (%) data.

C. reinhardtii (78.12%) and *A. platensis* (73.36%) as shown in Fig. 3. The statistical analysis in Fig. 4 illustrates the phosphate remediation efficiencies of the algae species.

It was concluded that by the end of the ten-day period, regardless of the algae species, a range of 73-85% of the initial wastewater phosphate content was removed in each of the reactors.

CONCLUSION

Based on the observations and test results, I conclude that all the algae specimens are able to remediate nitrate and phosphate content from wastewater. *Chlamydomonas reinhardtii* was shown to be most effective for nitrate removal, remediating up to 96% of the concentration and making the water safe for the environment. While all four specimens were effective, *Scenedesmus quadricauda* was shown to be the most effective for phosphate remediation of wastewater, remediating up to 85.22% of the phosphate and making the water able to be discharged into the environment. This experiment concludes that algae bioremediation of wastewater provides an effective and environmentally acceptable option for wastewater treatment.

This study was focused on nitrate and phosphate removal aspect of wastewater bioremediation. There are many other characteristics of wastewater treatment such as BOD and COD remediation and removal of toxic heavy metals, which can be studied further as future goals of this project.

MATERIALS AND METHODS

Collection of Wastewater and Algae Specimen: Due to regulations on handling wastewater, this study's wastewater was simulated by mixing fertilizer ammonium nitrate and diammonium phosphate with distilled water and filtering using Whatman No.1 filter paper to remove suspended solids. There are many different species of algae available, and for this study, species that are readily available and easy to grow

were collected. These species do not have adverse effects on environment or humans. They are also known to be pollution tolerant and have a high absorption capacity. Such species are considered to be effective in remediation of wastewater and so *Chlamydomonas reinhardtii*, *Arthrospira platensis* (*Spirulina*), *Scenedesmus quadricauda*, and *Chlorella vulgaris* were chosen for this study.

Experiment Setup: Algae need a healthy environment to grow and bioremediate, so a bioreactor system was designed using 500 ml Erlenmeyer flasks connected with an air pump to provide a source of carbon dioxide in the flasks. A fluorescent light was kept on for fast photosynthesis of green algae. The bioreactor was used as an experimental prototype of the Algae-based Wastewater Treatment System (AWTS). The filtered, untreated wastewater was used as the control solution. The algae solutions were made by adding 4 ml of the algae and 200 ml of filtered wastewater in a bioreactor flask. The test was conducted under controlled conditions (temperature $72 \pm 2^\circ\text{F}$) for a total duration of ten days, repeated in five different trials. Using a portable Hach DR-890 colorimeter, measurements of the nitrate and phosphate concentration were taken every day for ten days.³

The nitrate concentration of the solution was measured by cadmium reduction method (Hach Method 8039⁴). In this method, the cadmium metal reduces nitrates present in the sample to nitrite. This nitrite forms an amber-colored product by reacting in an acidic medium with sulfanilic acid. This amber color indicates the presence of nitrate and the intensity is converted to (N-NO₃) mg/L or ppm.

Similarly, the phosphate concentration of the solution was measured by the molybdate-ascorbic acid method (Hach Method 8048⁴). The orthophosphate present in the sample reacts with molybdate in the acid medium to produce a phospho-molybdate complex. Ascorbic acid reduces this complex and forms a blue color. This blue color indicates the presence of phosphate and the intensity is converted to (P-PO₄) mg/L or ppm.

Calculations of Remediation Efficiency: The nitrate and phosphate remediation efficiency of the algal specimen was calculated by the formula $Ei = \{(Co - Ci) / Co\} \times 100$ where Co and Ci were the concentrations of contaminant at the start of experiment (day 0) and day i, respectively.

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■ AUTHORS

Hrishika Roychoudhury is a sophomore from Ardrey Kell High School in Charlotte, NC. She is the founder of non-profit StemmEdHelp that helps students in leaning and spreading awareness on STEM education. She is deeply passionate about science research and won many awards in science competitions. The World Science Foundation selected her as one of 30 World Science Scholars from all over the world in 2019. She is also ISEF 2019 finalist and National TSA Conference 2019 finalist. Miss Roychoudhury is also an Indian classical dancer and is among the top 75 female chess players of her age in USA.